Temperature-switchable thermal barriers against thermal runaway propagation

A one- and two-dimensional finite-volume study

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github.com/adityameenak/switchable-barrier-runaway

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Abstract

A thermal barrier between lithium-ion cells faces a conflict. A conductive barrier spreads heat to the cooling system in normal operation but carries a runaway cell's heat to its neighbours. An insulating barrier blocks propagation but traps heat during normal use. A barrier whose conductivity drops sharply above a switch temperature could, in principle, do both. This report quantifies that tradeoff with a one-dimensional transient finite-volume model of a five-cell stack, using single-step Arrhenius self-heating, a smooth reversible switch law $k(T)$ and convective cooling at the stack ends. With illustrative parameters, a 2 mm switchable barrier holds the normal-operation peak temperature at the conductive barrier's 38.0 °C (against 64.5 °C for an aerogel-like insulator) while confining runaway to the trigger cell, whereas a conductive barrier lets runaway spread to all five cells within about a minute. A sweep of 361 barrier designs shows that containment is governed by a single quantity, the hot-state barrier conductance k_{off}/L, which must stay below about $32\,\mathrm{W\,m^{-2}\,K^{-1}}$ for the chosen end cooling. A switch-temperature sweep finds a window of about 38–175 °C in which the switch both cools and protects. The solver conserves energy to round-off and is verified against analytical, ODE-solver and convergence tests. The convergence study also showed that onset-fitted Arrhenius kinetics, extrapolated to burning-cell temperatures, produce a reaction front too thin for any practical mesh to resolve; a rate ceiling restores mesh convergence. The model is then extended with tabulated $k(T)$, latent heat and a delayed, one-way switch, and six real barrier materials are compared using datasheet and literature properties. Static insulators either contain runaway only with a large normal-use penalty (silica aerogel, 79 °C at its 2.25 mm containment thickness; ceramic fibre, 66 °C at 4.25 mm) or need thick layers (mica, 6.25 mm), and a conductive gap pad and a paraffin/graphite phase-change composite do not contain it at any thickness up to 10 mm. A published graphene/microsphere thermal switch contains runaway at 3.5 mm while keeping the normal-use peak within 1 K of the ideal conductor, and is the best of the six under every sensitivity case tested. Cell and kinetic parameters remain order-of-magnitude placeholders, so the conclusions are qualitative. Finally, 452 filled-silicone formulations (particles, silver nanowires, nanoparticle–nanowire hybrids, endothermic fillers and expandable microspheres) are screened with an effective-medium model extended for fin-limited nanowires, percolating wire networks and microsphere expansion, and checked against measured nanowire composites. Static formulations need at least 5.6 mm to contain runaway, and 8.5 mm or more to keep the normal-use peak near 44 °C. The recommended formulation is silicone with 40 vol% expandable microspheres and 20 vol% $Al(OH)_3$, which contains runaway at 2.9 mm (3.5 mm in the worst case tested) at a 42 °C normal-use peak. Conductive nanowire networks raise cold-state conductivity but are counterproductive unless the expansion breaks them.

1 Introduction

Thermal runaway in a lithium-ion cell is a self-accelerating sequence of exothermic decomposition reactions that can drive the cell to several hundred degrees and lead to fire [1–3]. In a module, the heat released by one failing cell can trigger runaway in its neighbours, as multi-cell abuse experiments have shown repeatedly [4, 5]. Preventing this cell-to-cell propagation is a central goal of battery-pack safety design. The usual approach places a thermal barrier between cells, which creates a design conflict:

- a *conductive* barrier helps move the heat cells generate in normal operation toward the cooling system, but also passes a runaway cell's heat directly to its neighbour;
- an *insulating* barrier, for example an aerogel sheet, blocks propagation but raises cell temperatures in everyday use, which accelerates ageing.

Temperature-responsive barriers and thermal switches, whose conductivity changes with temperature, are an active research topic as a way out of this conflict [6, 7].

Related work. Thermal abuse of single cells is commonly modelled by coupling heat conduction to Arrhenius kinetics for a set of decomposition reactions, an approach established for cylindrical and prismatic cells [8], applied to abuse behaviour of high-power cells [9], extended to three dimensions [10] and combined with experiments [11]. Propagation models extend this to packs and modules, with lumped [12] and three-dimensional descriptions [13] and analyses of the heat-transfer paths between cells [14]. On the mitigation side, studies have examined phase-change composites [15], aerogel-based insulation [16], model-based optimisation of heat-absorbing interstitial barriers [17] and polymer barrier materials [18]. These barriers are static: their conductivity does not respond to the operating state. Thermal switches, devices whose conductance changes with temperature or another stimulus, are reviewed in general in [19] and for batteries in [7], and a temperature-responsive regulator for battery modules has been demonstrated experimentally [6].

This work. This report complements that work with a deliberately minimal model that isolates the conductivity-switching function of a barrier, and maps which switch properties are needed for containment without penalising normal operation. It asks five questions:

1. Does an idealised switchable barrier deliver both the conductor's normal-use temperature and the insulator's containment?
2. What hot-state conductivity and thickness does it need?
3. How should its switch temperature be chosen?
4. Which of six real barrier materials, with properties taken from datasheets and the literature, best balances containment against normal-use temperature?
5. Which filled-silicone formulation (fillers, nanoparticles, nanowires and their combinations) should be made and tested, and how robust is that choice?

The emphasis is on a transparent, verified model rather than quantitative prediction. The cell and kinetic parameters are illustrative, order-of-magnitude values and have not been fitted to any particular cell. The barrier properties in the material comparison (Section 5.5) are sourced, with every assumed value flagged.

2 Model

2.1 Geometry and governing equations

The stack is five cells of thickness $L_c = 10\,\mathrm{mm}$ separated by four barriers of thickness L (default $2\,\mathrm{mm}$), modelled in one dimension through the stack thickness. Temperature obeys transient conduction with a temperature-dependent conductivity and volumetric sources,

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k(T)\,\frac{\partial T}{\partial x}\right) + q_{\mathrm{rxn}} + q_{\mathrm{gen}}, \tag{1}$$

with convective boundary conditions at both ends of the stack,

$$-k\,\frac{\partial T}{\partial x}\bigg|_{x=0} = h\,(T_\infty - T), \qquad k\,\frac{\partial T}{\partial x}\bigg|_{x=L_{\mathrm{stack}}} = h\,(T_\infty - T). \tag{2}$$

2.2 Cell self-heating

Each cell node carries a lumped reaction progress $\alpha \in [0,1]$ governed by single-step kinetics, a simplification of the multi-reaction schemes used in detailed abuse models [8–10]:

$$\frac{\mathrm{d}\alpha}{\mathrm{d}t} = k_{\mathrm{eff}}(T)\,(1-\alpha), \qquad q_{\mathrm{rxn}} = \Delta H_v \frac{\mathrm{d}\alpha}{\mathrm{d}t}, \qquad \Delta H_v = \rho c_p\,\Delta T_{\mathrm{ad}}, \tag{3}$$

so that a cell reacting adiabatically heats by exactly ΔT_{ad}. The rate constant combines an Arrhenius term with a smooth ceiling,

$$k_{\mathrm{eff}} = \frac{k_{\mathrm{arr}}\,k_{\max}}{k_{\mathrm{arr}} + k_{\max}}, \qquad k_{\mathrm{arr}} = A\,e^{-E_a/RT}. \tag{4}$$

With $E_a = 169\,\mathrm{kJ\,mol^{-1}}$ and $A = 7.3\times10^{16}\,\mathrm{s^{-1}}$, k_{arr} is about $10^{-4}\,\mathrm{s^{-1}}$ at 150 °C (slow onset) and about $1\,\mathrm{s^{-1}}$ at 250 °C. The ceiling $k_{\max} = 1\,\mathrm{s^{-1}}$ leaves onset unchanged; its purpose is explained in Section 4.1.

2.3 Barrier designs

Four designs are compared: no barrier (cells in direct contact); a static conductor ($k = 5\,\mathrm{W\,m^{-1}\,K^{-1}}$); a static, aerogel-like insulator ($k = 0.03\,\mathrm{W\,m^{-1}\,K^{-1}}$); and a switchable barrier that conducts below a switch temperature T_{sw} and insulates above it. The switch law is a logistic blend in $\ln k$,

$$\ln k(T) = \ln k_{\mathrm{off}} + \frac{\ln(k_{\mathrm{on}}/k_{\mathrm{off}})}{1 + e^{(T-T_{\mathrm{sw}})/w}}, \tag{5}$$

with $k_{\mathrm{on}} = 5\,\mathrm{W\,m^{-1}\,K^{-1}}$, $k_{\mathrm{off}} = 0.05\,\mathrm{W\,m^{-1}\,K^{-1}}$, $T_{\mathrm{sw}} = 120\,^\circ\mathrm{C}$ and $w = 5\,\mathrm{K}$. The switch is reversible and local: every barrier node switches on its own temperature, so a barrier with a steep internal gradient can be insulating on its hot side and conducting on its cold side. Blending $\ln k$ rather than k matters for a 100× switch ratio. A linear blend leaves k several times k_{off} until about 20 K above T_{sw}, so the barrier would barely insulate near its nominal switch point. All barriers share the same volumetric heat capacity, which isolates the effect of conductivity.

2.4 Real barrier materials

Comparing real materials needs three extensions of the barrier model. Each reduces to the idealised designs when switched off, and all earlier results are unchanged.

- **Tabulated conductivity.** $k(T)$ is interpolated from published points, linearly in $\ln k$, and held constant beyond the first and last point.
- **Latent heat.** A phase-change barrier stores enthalpy

$$H(T) = \rho c_p (T - T_{\mathrm{ref}}) + \rho L \, \Phi\!\left(\frac{T - T_m}{\sigma_m}\right), \tag{6}$$

  with latent heat L released over a Gaussian melting range (Φ is the normal cumulative distribution, $\sigma_m = 3\,\mathrm{K}$).
- **Switch dynamics.** The conducting fraction s of a switch (the exponent in Eq. (5)) relaxes toward its equilibrium value $s_{\mathrm{eq}}(T)$ with time constant τ, $\mathrm{d}s/\mathrm{d}t = (s_{\mathrm{eq}} - s)/\tau$. A one-way switch can only move toward the insulating state, $s^{n+1} = \min(s^n, \cdot)$.

Six candidates were chosen to cover the main barrier mechanisms (Table 1): three static insulators (silica aerogel blanket, phlogopite mica paper, alumina–silica ceramic fibre), a static conductor (silicone gap pad), a phase-change composite (paraffin with 10 wt% expanded graphite) and a thermal switch (the graphene/expandable-microsphere regulator of Wang et al. [6]). Every value is taken from a manufacturer datasheet or a peer-reviewed paper, except where marked as assumed. The data and their provenance are kept in `runaway/candidates.py`, which also records how each material is mapped onto the model.

Table 1: Barrier candidates. k is through-plane, at the temperatures shown. [a]Assumed generic value for the material class (not in the source). [b]From a search summary of the source, not yet checked against the original.

Material (source)	k [$\mathrm{W\,m^{-1}K^{-1}}$] at T [°C]	ρ [$\mathrm{kg\,m^{-3}}$]	c_p [$\mathrm{J\,kg^{-1}K^{-1}}$]	Other
Silica aerogel [20]	0.020 @ 0, 0.028 @ 200, 0.046 @ 400, 0.089 @ 600	200	1000[a]	max. 650 °C
Mica paper [21]	0.23 @ 50	1900	850[a]	max. 700 °C
Ceramic fibre [22]	0.06 @ 260, 0.13 @ 538, 0.23 @ 816	128	1000[a]	max. 1177 °C
Silicone gap pad [23]	3.0[b] @ 25	3200[b]	1000[b]	max. 200 °C[b]
Paraffin/EG PCM [24]	9.99 @ 25[a]	1103	2000[a]	$L = 240\,\mathrm{kJ\,kg^{-1}}$, $T_m \approx 74\,°\mathrm{C}$
Graphene switch [6]	1.33 @ 25, 0.10 @ ∼100	1000[a]	1000[a]	switches within 30 s

Where the data do not determine a model input, the choice is stated and later varied (Table 6). The graphene switch is modelled with $k_{\mathrm{on}} = 1.33$, $k_{\mathrm{off}} = 0.10\,\mathrm{W\,m^{-1}K^{-1}}$, $T_{\mathrm{sw}} = 85\,°\mathrm{C}$ and $w = 5\,\mathrm{K}$, so that k reaches about $0.11\,\mathrm{W\,m^{-1}K^{-1}}$ at 100 °C, and its 30 s response as a lag with $\tau = 10\,\mathrm{s}$. Expanding microspheres suggest a one-way switch, which is the default. The gap pad and the PCM are given constant k. Several of these products are not made for battery barriers: the aerogel and ceramic fibre data are for industrial blankets, and the PCM was developed for spacecraft electronics. The gap pad is rated only to 200 °C and the PCM was tested only to 85 °C, so their behaviour at runaway temperatures is an extrapolation.

2.5 Filled-silicone compositions

Many practical barriers and gap fillers are polymers loaded with particles. To compare compositions rather than finished products, a silicone elastomer matrix [25] ($k_m = 0.20\,\mathrm{W\,m^{-1}K^{-1}}$, $\rho = 1030\,\mathrm{kg\,m^{-3}}$) is combined with eight fillers (Table 2): calcium carbonate as nanoparticles and microparticles, the flame-retardant hydroxides $Al(OH)_3$ (ATH) and $Mg(OH)_2$ (MDH), hexagonal boron nitride platelets, silver nanowires, multi-walled carbon nanotubes, and hollow glass microspheres.

Effective conductivity. The through-plane conductivity of the composite is estimated with the effective-medium model of Nan et al. [26] for randomly oriented spheroidal inclusions with an interfacial (Kapitza) resistance R_K. For filler volume fractions f_i,

$$\frac{k}{k_m} = \frac{3 + \sum_i f_i \left[2\beta_{11}(1 - L_{11}) + \beta_{33}(1 - L_{33})\right]}{3 - \sum_i f_i \left[2\beta_{11} L_{11} + \beta_{33} L_{33}\right]}, \qquad \beta_{jj} = \frac{k^c_{jj} - k_m}{k_m + L_{jj}(k^c_{jj} - k_m)}, \tag{7}$$

where L_{jj} are the depolarisation factors of a spheroid with aspect ratio $p = a_3/a_1$, and $k^c_{jj} = k_{p,jj}/(1+\gamma L_{jj} k_{p,jj}/k_m)$ is the particle conductivity reduced by the interface, with $\gamma = (2+1/p)\,a_K/a_1$ for rods ($p \geq 1$), $\gamma = (1 + 2p)\,a_K/a_3$ for platelets ($p \leq 1$) and Kapitza radius $a_K = R_K k_m$. For spheres Eq. (7) reduces to the Hasselman–Johnson result and, with $R_K = 0$, to Maxwell's. Particle size enters only through a_K/a: when the Kapitza radius approaches the particle size, the interface dominates, which is why nanoparticles conduct less than microparticles of the same material. The model neglects percolating filler networks, so it is used only below each shape's percolation range (40 vol% for spheres, 30 % for platelets, 5 % for wires and tubes). For the very high aspect ratios of nanowires and nanotubes it is an upper bound: contact resistance, waviness and agglomeration make measured composites conduct less.

Endothermic fillers. ATH, MDH and $CaCO_3$ decompose on heating, releasing water or CO_2 and absorbing heat. Each is represented with the latent-heat mechanism of Eq. (6), carrying the filler's mass fraction of its decomposition enthalpy over a Gaussian temperature range. The gas release, and the irreversibility of the reaction, are not modelled. Density and heat capacity follow volume- and mass-weighted mixture rules.

Table 2: Fillers. Particle shapes and sizes are representative choices (design variables), not properties of a specific product. $R_K = 10^{-8}\,\mathrm{m^2K\,W^{-1}}$ (assumed) unless stated. [a]Assumed. [b]From a summary of the source.

Filler	k_p [$\mathrm{W\,m^{-1}K^{-1}}$], shape	ρ [$\mathrm{kg\,m^{-3}}$]	Decomposition
$CaCO_3$, 50 nm / 5 μm	3.5[b] [27], spheres	2710 [28]	1.78 $\mathrm{MJ\,kg^{-1}}$, $700 \pm 40\,^\circ$C [29]
$Al(OH)_3$ (ATH), 1 μm	3[a], spheres	2340 [28]	1.17 $\mathrm{MJ\,kg^{-1}}$, $260 \pm 20\,^\circ$C[b] [30]
$Mg(OH)_2$ (MDH), 1 μm	3[a], spheres	2390 [28]	1.36 $\mathrm{MJ\,kg^{-1}}$, $365 \pm 20\,^\circ$C[b] [30]
h-BN, 5 μm × 100 nm	300 in-plane, 3 through[b] [31], platelets ($p = 0.02$)	2100[a]	–
Ag nanowires, 84 nm × 20 μm	332[b] [32], rods ($p \approx 240$)	10490	–
MWCNT, 20 nm × 10 μm	3000 axial [33], rods ($p = 500$), $R_K = 8.3 \times 10^{-8}$ [34]	1800[a]	–
Hollow glass bubbles (3M K1)	0.047 [35], spheres, 65 μm	125 [35]	–

Eleven compositions are compared: neat silicone; single fillers at a high but model-valid loading ($CaCO_3$ 30 vol% as nano- and microparticles, ATH and MDH 40 %, h-BN 30 %, Ag nanowires and CNTs 5 %, glass bubbles 50 %); and two hybrids that pair an insulating or a conducting filler with an endothermic one (40 % glass bubbles + 20 % ATH; 20 % h-BN + 20 % ATH). Figure 1 shows the

resulting conductivities. Silicone degrades above roughly 200–300 °C, so, like the gap pad, all these composites are extrapolated beyond their service range in the abuse scenario.

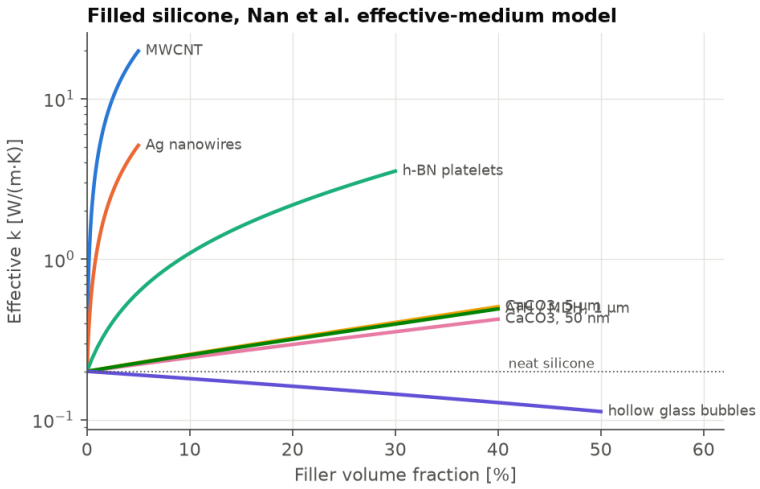


Figure 1: Effective conductivity of silicone with each filler of Table 2 (Nan et al. model). The nanowire and nanotube curves are upper bounds; a fin-limited model for wires is introduced in Section 2.6.

2.6 Formulation design: nanowires, hybrids and an expanding switch

The eleven fixed compositions sample single fillers. Designing a formulation needs the model to answer three further questions: how much do nanowires really add, what happens when nanoparticles and nanowires are combined, and can a composite be made to switch? Four extensions, implemented in `runaway/formulation.py`, address these. Each reduces to Eq. (7) when its ingredient is absent.

Fin-limited nanowires. Applied to a nanowire, Eq. (7) lets the wire carry heat along its whole length. For aspect ratios of 100–1000 this predicts about $5\,\mathrm{W\,m^{-1}K^{-1}}$ for 5 vol% silver nanowires in silicone (Fig. 1), several times what is measured in comparable composites. Heat has to enter and leave a wire through its surface, across the interface resistance and the polymer around it. The wire therefore behaves as a fin with exchange coefficient $h_{\mathrm{eff}} = [R_K + D/(2k_h)]^{-1}$, where D is the wire diameter and k_h the conductivity of the host it sits in. Its effective axial conductivity is

$$k_{\mathrm{ax}} = \eta\, k_w, \qquad \eta = \frac{\tanh(m\ell)}{m\ell}, \qquad m = \sqrt{\frac{4h_{\mathrm{eff}}}{k_w D}}, \tag{8}$$

with ℓ half the wire length. k_{ax} replaces $k_{p,33}$ in Eq. (7). For a $40\,\mathrm{nm} \times 40\,\mu\mathrm{m}$ wire in silicone, $\eta \approx 0.03$. The intrinsic wire conductivity falls with diameter because of boundary scattering. It is taken as $k_w = 429/(1 + D_0/D)\,\mathrm{W\,m^{-1}K^{-1}}$ with $D_0 = 24.5\,\mathrm{nm}$, which reproduces the measured $332\,\mathrm{W\,m^{-1}K^{-1}}$ at 84 nm [32] and bulk silver for thick wires (an assumed form).

Percolating wire networks. Randomly oriented slender rods first form a connected network when the number density times their mean excluded volume reaches one [36, 37]. For rods of length L and diameter D this gives the threshold $\phi_c = D/(2L)$, about 0.05 vol% for $40\,\mathrm{nm} \times 40\,\mu\mathrm{m}$ wires. Above it, heat can also cross wire–wire junctions. Treating each wire as isothermal and summing the heat dissipated over the junctions in a uniform gradient gives $k_{\mathrm{net}} = (\pi/72)\, n^2 L^4 D\, G_c$ for n

wires per unit volume and junction conductance G_c, the n^2 scaling found for nanotube networks [38]. With $G_c = g\,k_h D$ (a junction bridged by polymer, g a dimensionless junction coefficient) and the 3D conductivity exponent of two near the threshold,

$$k_{\text{net}} = \frac{2}{9\pi}\left(\frac{L}{D}\right)^2 g\,k_h\,(\phi_w - \phi_c)^2, \tag{9}$$

which is added in parallel to the dispersed-wire result. Polymer composites show little or no thermal percolation in practice [39], so g is small by default ($g = 0.05$) and is varied from 0 to 0.5, with a further $\times 10$ for sintered junctions.

Nanoparticle–nanowire hybrids. Particles and wires are combined by homogenising in two steps. Spherical or platelet particles are first mixed into the matrix with Eq. (7) (fractions taken relative to the non-wire volume), giving a host conductivity k_h. The wires are then embedded in that host. Particles help wires in two ways: a better host raises h_{eff} in Eq. (8), and it raises the junction conductance in Eq. (9).

Expandable-microsphere switch. Thermoplastic microspheres filled with a volatile hydrocarbon expand irreversibly when heated through a supplier-specified range. For the grade used here (Expancel 051 DU 40) expansion begins at 105–110 °C and peaks at 142–151 °C [40]. A volume fraction ϕ_s of spheres expanding by a volume ratio E dilutes every other phase by $X = 1 + \phi_s(E-1)$ and turns the spheres into gas cells occupying $\phi_s E/X$ ($k = 0.04\,\mathrm{W\,m^{-1}K^{-1}}$, assumed). The hot-state conductivity k_{off} is the two-step model evaluated with the diluted fractions. If the dilution takes the wire fraction below ϕ_c the network term vanishes, so wires can amplify the switch. The opposite, conservative assumption keeps the network intact in the polymer struts between the gas cells ("segregated"). Free expansion exceeds 40-fold (the datasheet's expanded density is at most $25\,\mathrm{kg\,m^{-3}}$), but a cured elastomer resists it, so $E = 5$ is assumed and varied from 2 to 20. The composite enters the solver as a one-way switch of the kind in Section 2.4: k_{on} and k_{off} as computed, $T_{\text{sw}} = 127\,^\circ\mathrm{C}$ at the middle of the expansion range, a width such that 10–90 % of the change spans T_{start} to T_{max} ($w = 8.9\,\mathrm{K}$), and $\tau = 10\,\mathrm{s}$ as for the graphene switch. Barrier thickness and heat capacity are held at their unexpanded values. A barrier that also grew thicker would insulate better, so this is conservative for containment. ATH or MDH added to the formulation carries its decomposition enthalpy as in Section 2.5. Alumina particles ($k = 30\,\mathrm{W\,m^{-1}K^{-1}}$ [41]) are added to the filler set as nanoparticles (50 nm) and microparticles (5 μm).

Check against measured composites. Figure 2a compares the nanowire model with measured silver-nanowire/epoxy composites. Without fitting, the default interface resistance ($R_K = 10^{-8}\,\mathrm{m^2K\,W^{-1}}$) gives $1.11\,\mathrm{W\,m^{-1}K^{-1}}$ for 4 vol% silica-coated wires, against 1.03 measured [42] (the wire size is not in the abstract and 100 nm × 15 μm is assumed). It gives $0.33\,\mathrm{W\,m^{-1}K^{-1}}$ for 0.46 vol% of 120 nm × 15 μm wires, against 0.35 [43]. The same study's uncoated wires reached only $0.57\,\mathrm{W\,m^{-1}K^{-1}}$ [42]. Reproducing that needs an effective interface resistance 350 times larger, $3.5 \times 10^{-6}\,\mathrm{m^2K\,W^{-1}}$, which is read here as poor dispersion and used as a sensitivity case. For hybrids the model is conservative. With 40 vol% alumina in epoxy, adding 0.5 vol% nanowires raised the measured conductivity by 106 % [44]. The two-step model predicts 13 %, because it does not represent wires bridging neighbouring particles. Per unit volume, nanowires raise k about ten times more than particles (Fig. 2b), but the fin limit keeps 5 vol% of 84 nm wires in silicone at about $1\,\mathrm{W\,m^{-1}K^{-1}}$, not the 5 of Fig. 1.

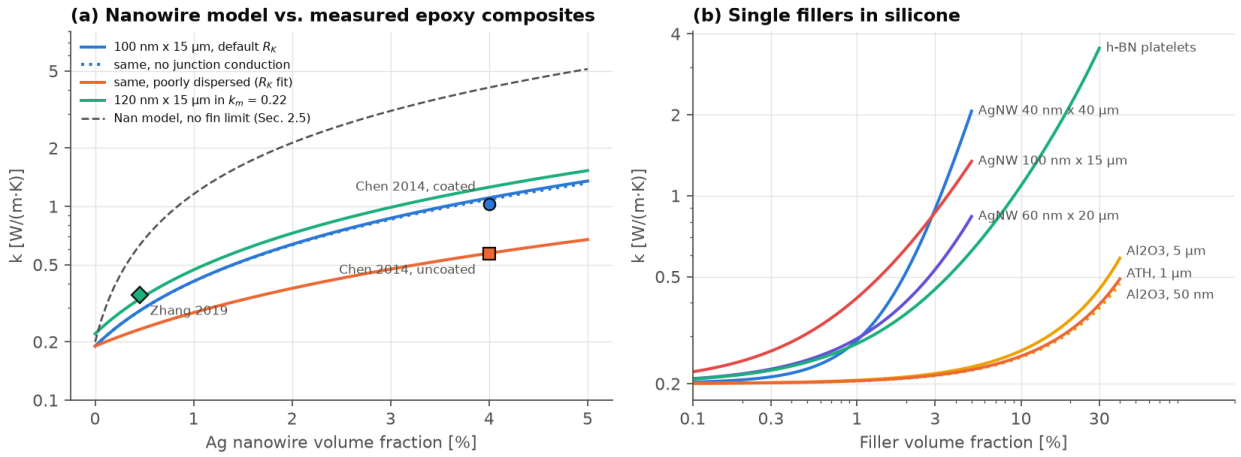


Figure 2: (a) Fin-limited nanowire model (lines) against measured silver-nanowire/epoxy composites (markers); the dashed line is the unmodified Nan model. (b) Single fillers in silicone with the formulation model. Below the percolation threshold the nanowire curves are set by the fin limit; above it, the thin, long wires gain a junction network.

2.7 Two-dimensional model with a cold plate

In the 1D model the only heat sink is the pair of stack ends, which exaggerates how much an insulating barrier heats the pack in normal use. Real modules stand on a liquid-cooled plate. The 2D model adds the cell height y ($H = 100\,\mathrm{mm}$):

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_y \frac{\partial T}{\partial y}\right) + q_{\mathrm{rxn}} + q_{\mathrm{gen}}, \tag{10}$$

with the 1D end condition at $x = 0, L$; a cold plate at $y = 0$ ($h_{\mathrm{plate}} = 200\,\mathrm{W\,m^{-2}K^{-1}}$ including the interface material, coolant at 25 °C) under both cells and barriers; and natural convection ($h = 5\,\mathrm{W\,m^{-2}K^{-1}}$) at the top. Cells are anisotropic: $k_x = 0.8$ is the through-plane value of the 1D model and $k_y = 20\,\mathrm{W\,m^{-1}K^{-1}}$ the in-plane value along the electrode sheets. Barriers are isotropic. These values, like the cell parameters, are illustrative. The trigger condition, kinetics and propagation criterion are those of the 1D model, with the cell mean taken over its area.

2.8 Scenarios and parameters

Abuse. Cell 1 starts at 300 °C and everything else at 25 °C. A cell is counted as *propagated* when its volume-mean α exceeds 0.9.

Normal operation. Every cell generates a uniform $q_{\mathrm{gen}} = 2 \times 10^4\,\mathrm{W\,m^{-3}}$, the reaction is switched off (its rate is negligible below 70 °C), and Eq. (1) is solved to steady state. The quantity of interest is the peak cell temperature.

Parameters. Table 3 lists the parameters. Two were changed from the initial estimates during development. First, the end cooling coefficient was raised to $h = 100\,\mathrm{W\,m^{-2}\,K^{-1}}$, representing an end plate connected to a cooling system. With natural-convection values ($h \lesssim 30\,\mathrm{W\,m^{-2}\,K^{-1}}$) no barrier, not even the insulator, prevented propagation in this 1D model, because the stack ends are the only heat sink and the trigger cell's roughly $20\,\mathrm{MJ\,m^{-2}}$ must eventually leave somewhere. Second, the rate ceiling $k_{\max}$ was added (Section 4.1).

Table 3: Model parameters. All values are illustrative order-of-magnitude placeholders and have not been fitted to a specific cell or material.

Quantity	Value	Note
Cell k, ρ, c_p	$0.8\,\mathrm{W\,m^{-1}K^{-1}}$, $2500\,\mathrm{kg\,m^{-3}}$, $1000\,\mathrm{J\,kg^{-1}K^{-1}}$	through-plane, homogenised
E_a, A	$169\,\mathrm{kJ\,mol^{-1}}$, $7.3\times10^{16}\,\mathrm{s^{-1}}$	single-step kinetics
ΔT_{ad}, $k_{\max}$	$600\,\mathrm{K}$, $1\,\mathrm{s^{-1}}$	
Conductor / insulator k	5 / $0.03\,\mathrm{W\,m^{-1}K^{-1}}$	static
Switch k_{on}, k_{off}, T_{sw}, w	5 and $0.05\,\mathrm{W\,m^{-1}K^{-1}}$, 120 °C, 5 K	reversible
Barrier ρc_p, thickness	$10^6\,\mathrm{J\,m^{-3}K^{-1}}$, 2 mm	all designs
h, T_∞	$100\,\mathrm{W\,m^{-2}K^{-1}}$, 25 °C	both ends
q_{gen} (normal operation)	$2\times10^4\,\mathrm{W\,m^{-3}}$	cells only

3 Numerical method

Spatial discretisation. Each layer is meshed separately (40 volumes per cell, 12 per barrier), so every material interface lies on a control-volume face. The face conductance is the series resistance of the two adjoining half-volumes,

$$G_{i+1/2} = \left(\frac{\Delta x_i}{2k_i} + \frac{\Delta x_{i+1}}{2k_{i+1}} \right)^{-1}, \tag{11}$$

which is the distance-weighted harmonic mean of k [45]. It is exact for steady conduction through layers in series and handles the 27× jump at an insulator–cell interface without special treatment. Boundary faces add $1/h$ in series.

Time integration. Reaction and conduction are coupled by first-order (Lie) operator splitting [46]. In the *reaction step*, Eq. (3) is linear in α for a frozen rate constant and has the exact solution

$$\alpha^{n+1} = 1 - (1-\alpha^n)\, e^{-k_{\mathrm{eff}}\Delta t}, \tag{12}$$

where k_{eff} is evaluated at a predicted mid-step temperature (an exponential midpoint rule). The released heat is added as $\Delta T_{\mathrm{ad}}\,(\alpha^{n+1}-\alpha^n)$. The update is stable for any Δt, keeps $0 \le \alpha \le 1$ and conserves energy exactly, which makes the stiff kinetics tractable. In the *conduction step*, backward Euler is used with $k(T)$ lagged from the start of the step (Picard refinement is available). The resulting tridiagonal system is solved with a banded direct solver.

Step control. The time step is

$$\Delta t = \min\left(\Delta t_{\max},\ \frac{\Delta T_{\mathrm{rxn,max}}}{\max_i \dot{T}_{\mathrm{rxn},i}} \right), \tag{13}$$

with $\Delta T_{\mathrm{rxn,max}} = 2\,\mathrm{K}$ and $\Delta t_{\max} = 0.5\,\mathrm{s}$. This is an accuracy limit rather than a stability limit: it shrinks the step only while some node is igniting.

Steady state. Normal operation is solved by pseudo-transient continuation: backward-Euler steps from ambient with a geometrically growing step. For the switchable barrier this follows the physical start-up path, so if self-heating trips the switch, the solution lands on the hot (insulating) branch.

Latent heat. With a phase-change barrier, the heat capacity in the backward-Euler step is replaced by the chord capacity $C^* = [H(T^{n+1}) - H(T^n)]/(T^{n+1} - T^n)$ from Eq. (6), and the step is iterated to a tolerance of 10^{-7} K. The stored-energy change then equals the enthalpy change exactly, so a node that crosses the whole melting range in one step cannot skip the latent heat, which an apparent-heat-capacity method evaluated at the old temperature would do.

Two dimensions. The 2D solver uses the same scheme on a tensor-product mesh (10 volumes per cell and 4 per barrier in x, 20 over the height), with harmonic-mean face conductances in both directions and a sparse direct solve per step. The coarser x mesh is checked in Section 4.

Energy accounting. The solver tracks stored enthalpy, unreleased chemical energy and cumulative boundary loss. Because both sub-steps are conservative, the discrete energy balance closes to round-off. The largest relative residual in any run reported here is 3×10^{-10} (a phase-change barrier, set by the iteration tolerance); without latent heat it is below 10^{-11}.

4 Verification

The test suite runs automatically on every commit and checks the following.

1. **Energy conservation.** With insulated ends and no reaction, total thermal energy stays constant to a relative 10^{-12} over 2000 s, with the nonlinear switchable barrier in place. With the reaction on, thermal plus chemical energy is conserved, and with convection the budget closes to below 10^{-10}.
2. **Series conduction.** For a cell | barrier | cell stack held between two different ambient temperatures, the steady solution matches the analytical series-resistance profile to 10^{-8} K, for both 6× and 27× conductivity jumps, on a deliberately coarse mesh. For uniform generation in a single cell, the peak matches $T_\infty + qL_c/(2h) + qL_c^2/(8k)$.
3. **Reaction.** A single adiabatic reacting node ends at exactly $T_0 + \Delta T_{\mathrm{ad}}$. Its time to 50 % conversion matches a tight-tolerance implicit ODE solution (Radau IIA [47]) to within 1 %, with and without the rate ceiling.
4. **Convergence.** The cell-2 propagation time converges under mesh refinement with an observed order above 1.5, and under refinement of $\Delta T_{\mathrm{rxn,max}}$ (Fig. 3).
5. **Material extensions.** Tabulated $k(T)$ reproduces the published points. In an insulated stack with a phase-change barrier, enthalpy is conserved to 10^{-9} and the final uniform temperature satisfies the enthalpy balance including melting. A one-way switch heated past T_{sw} stays insulating after cooling, while a reversible one recovers, and a larger τ delays switching.
6. **Composites.** For spheres the effective-medium model reproduces the Maxwell and Hasselman–Johnson formulas to 10^{-12}; depolarisation factors satisfy $2L_{11} + L_{33} = 1$ and the rod and disc limits; and mixture rules conserve mass and latent heat.
7. **Formulations.** Without wires or microspheres the two-step model equals the Nan model to 10^{-12}; the dilute limit matches Maxwell's first-order result; the percolation threshold and the quadratic growth above it follow Eq. (9); a short wire is nearly isothermal ($\eta > 0.95$); the two calibration points are reproduced to within 10 %; and expansion lowers k, more so for larger E, with the segregated network never insulating better than the dilute one.
8. **2D solver.** With the cold plate and top removed, the 2D solver reproduces the 1D propagation times and normal-operation peak to 10^{-9}. For a single uniformly heated cell cooled only by

the plate it matches the analytical parabolic profile up to the expected $q\Delta y^2/8k_y$ boundary term, and the energy budget with plate cooling and latent heat closes to below 10^{-9}.

4.1 Why a rate ceiling is needed

The kinetic parameters describe the onset region (150–250 °C). Extrapolated to a burning cell near 900 °C, pure Arrhenius kinetics give $k_{\mathrm{arr}} \sim 10^9\,\mathrm{s}^{-1}$, which implies a reaction front inside the cell of order $\sqrt{\kappa/k_{\mathrm{arr}}} \sim 10^{-7}$ m, where $\kappa = k/\rho c_p$ is the thermal diffusivity. No practical mesh resolves this, so the front speed is set by the mesh rather than by the physics. The convergence test exposed the problem. Without the ceiling, the cell-2 propagation time drifts from 14.3 to 10.9, 11.1 and 8.8 s as the mesh is refined from 10 to 80 volumes per cell, with no sign of settling (Fig. 3).

Real high-temperature decomposition is limited by transport and multi-step chemistry, not by an ever-faster single step. The smooth ceiling in Eq. (4) with $k_{\max} = 1\,\mathrm{s}^{-1}$ makes a single cell burn over a few seconds, broadens the front to about $\sqrt{\kappa/k_{\max}} \approx 0.6$ mm and restores convergence. The default mesh is within 2 % of a 4× finer one, and halving $\Delta T_{\mathrm{rxn,max}}$ changes the result by 0.1 %. The contained cases are also converged: the peak temperature in cell 2 behind the switchable barrier varies by less than 0.1 K over a 100× range of $\Delta t_{\max}$, and by about 1 K from 20 to 80 volumes per cell.

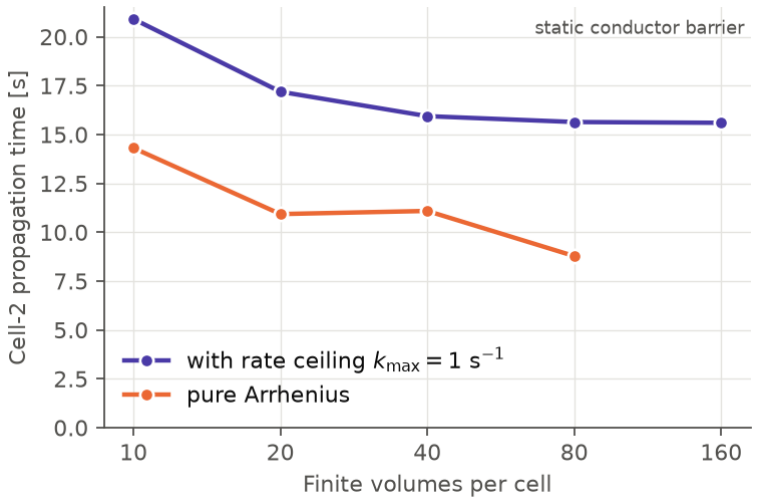


Figure 3: Mesh convergence of the cell-2 propagation time for the static-conductor stack. With pure Arrhenius kinetics the result does not converge, because the reaction front is unresolvably thin. With the rate ceiling it converges at roughly second order.

5 Results

5.1 The tradeoff

Table 4 and Fig. 4 summarise the four designs. Without a barrier or with the conductor, runaway spreads to every cell: each neighbour ignites 14–16 s after the previous one, and the whole stack has reacted within about one minute (Fig. 5). The insulator confines runaway to the trigger cell but raises the normal-operation peak by 26.5 K. The switchable barrier gives the conductor's 38.0 °C in normal use *and* the insulator's containment.

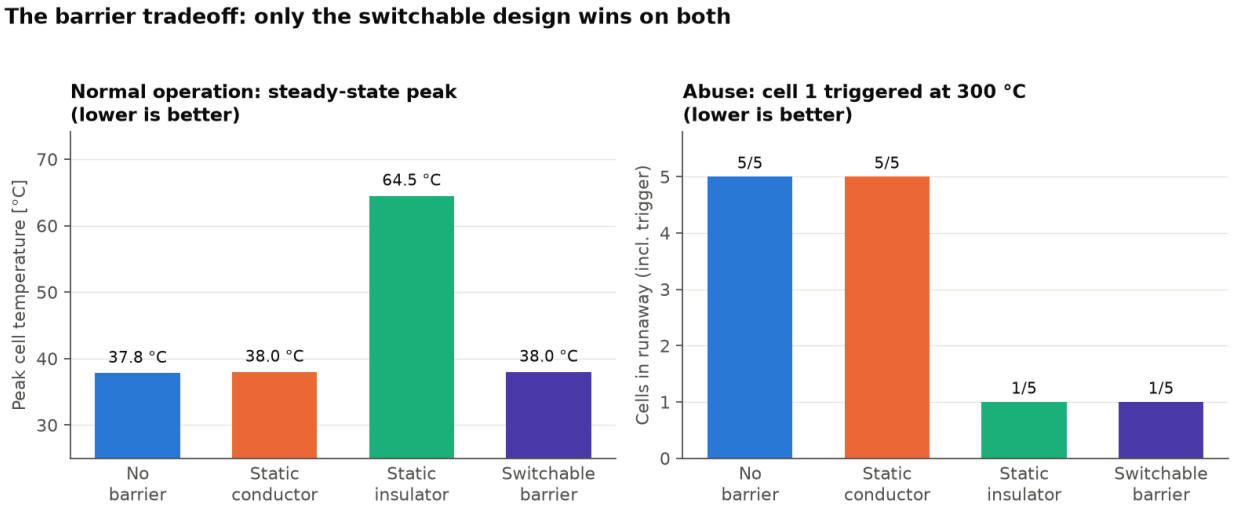


Figure 4: Normal-operation steady-state peak temperature (left) and number of cells in runaway after triggering cell 1 (right) for each design.

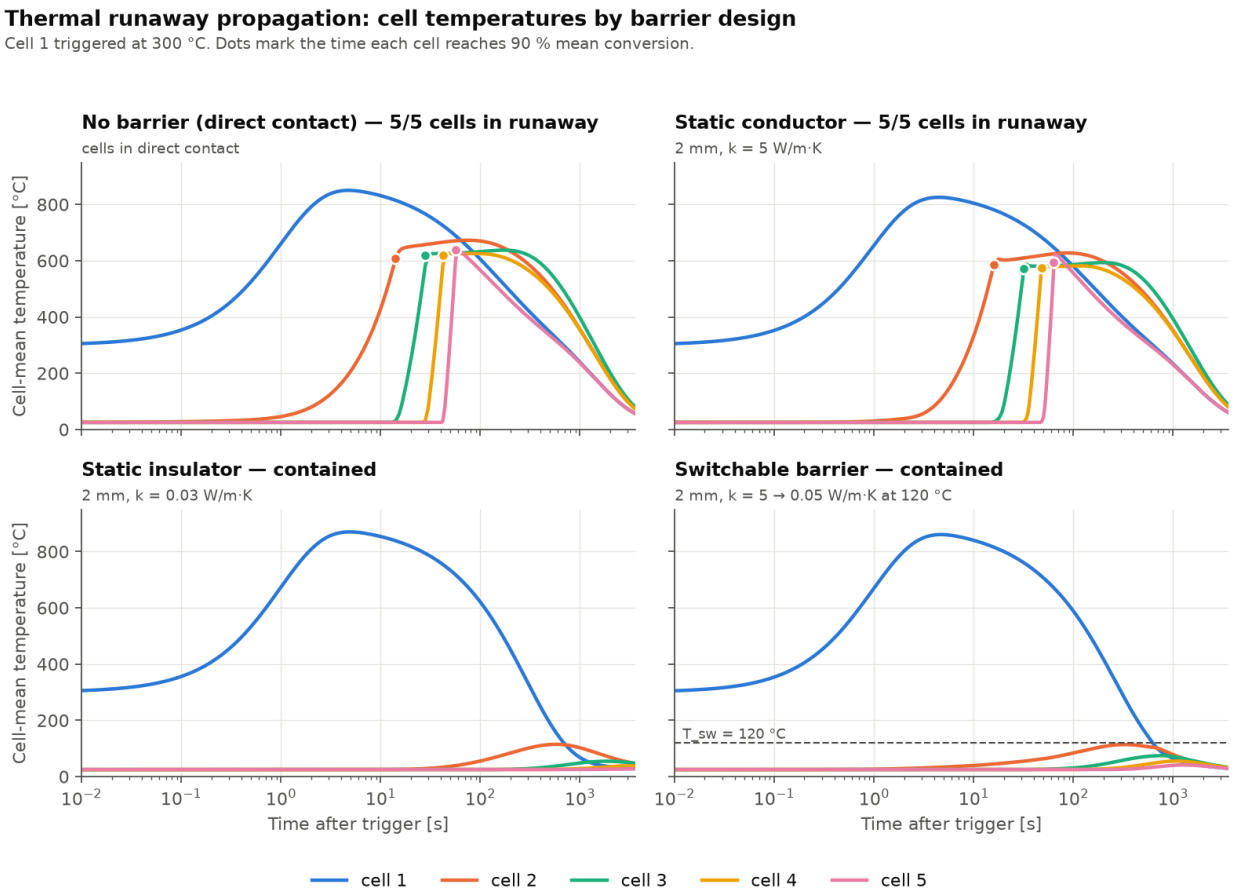


Figure 5: Cell-mean temperatures after triggering cell 1, one panel per barrier design. Dots mark the time each cell reaches 90 % mean conversion.

Table 4: Outcomes for the four designs (2 mm barriers where present).

Design	Normal-op. peak	Cells in runaway	Propagation times, cells 1–5 [s]
None (direct contact)	37.8 °C	5/5	2.3, 14.1, 28.2, 42.2, 56.1
Static conductor	38.0 °C	5/5	2.3, 15.9, 31.8, 47.8, 63.6
Static insulator	64.5 °C	1/5	2.3, –, –, –, –
Switchable	38.0 °C	1/5	2.3, –, –, –, –

The trigger cell peaks at 850–870 °C, slightly below $300 + \Delta T_{\mathrm{ad}} = 900$ °C, because it loses heat while it reacts. Propagated cells settle near 25 + 600 °C plus whatever preheating they received. Both values are consistent with the energy content of the model.

5.2 Mechanism of containment

Figure 6 shows how the switch works. The hot face of the first barrier switches off and holds a temperature drop of roughly 700 K across a few millimetres while cell 1 sheds its heat through the cooled end. The cold side of the same barrier stays below T_{sw} and keeps conducting, as do the downstream barriers, so cell 2 spreads whatever heat leaks through into cells 3–5. One consequence is easy to miss. Cell 2 reaches a higher local peak behind the switchable barrier (153 °C, 3 % conversion) than behind the static insulator (124 °C, 0.2 %), because only part of the switchable barrier's thickness is in the insulating state.

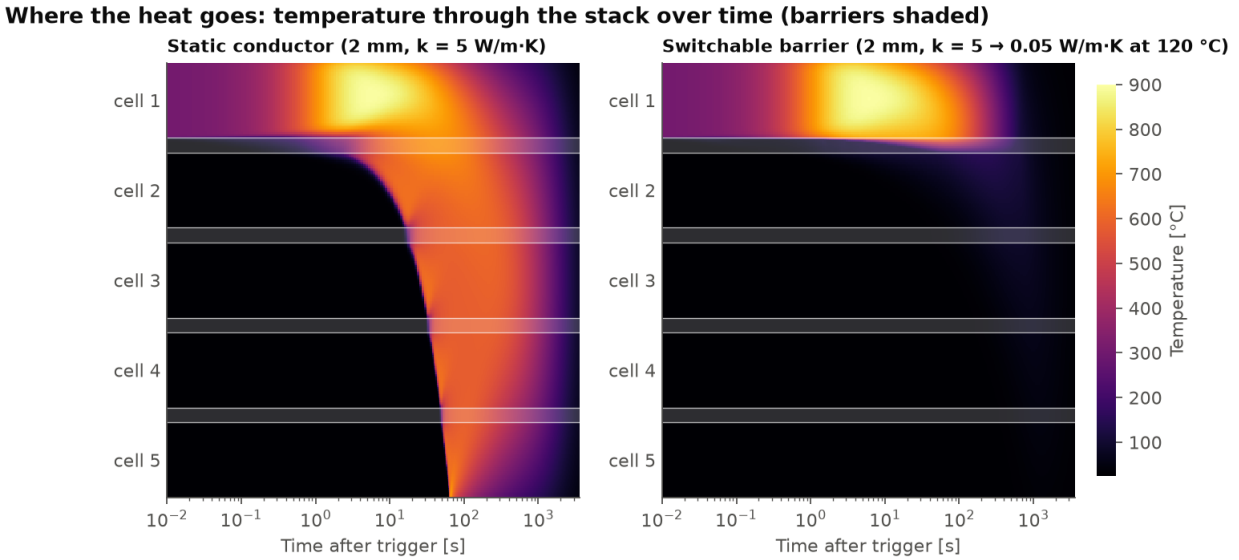


Figure 6: Temperature through the stack over time for the conductor (left) and the switchable barrier (right). Barriers are shaded.

5.3 Design map

Figure 7 varies the switchable barrier's thickness L (0.5–5 mm) and hot-state conductivity k_{off} (0.02–1 $\mathrm{W\,m^{-1}K^{-1}}$) over 361 simulations. Two features stand out. First, the outcomes are binary: every run ends with either one or all five cells in runaway. Once cell 2 ignites, each later cell faces a trigger at least as severe, so the barrier must stop the first hand-off. Second, the boundary follows a line of constant hot-state conductance, $k_{\mathrm{off}}/L \approx 32\,\mathrm{W\,m^{-2}K^{-1}}$, about a third of the end-cooling

coefficient. Physically, the runaway cell must lose heat to the cooled end faster than it leaks into its neighbour. The default design ($k_{\mathrm{off}}/L = 25\,\mathrm{W\,m^{-2}K^{-1}}$) lies about 20 % inside the boundary, a modest margin.

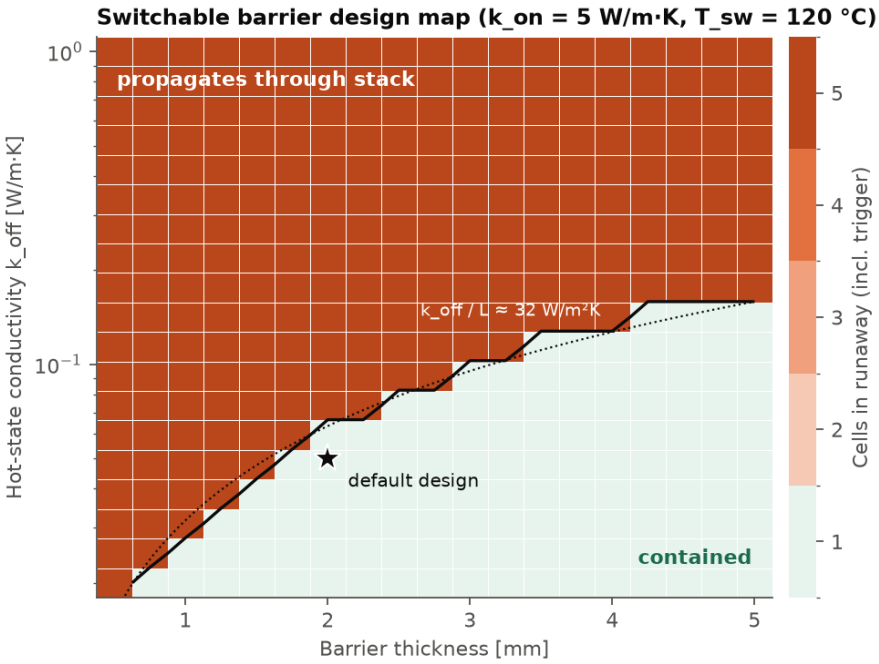


Figure 7: Cells in runaway over barrier thickness and hot-state conductivity ($k_{\mathrm{on}} = 5\,\mathrm{W\,m^{-1}K^{-1}}$, $T_{\mathrm{sw}} = 120\,^{\circ}\mathrm{C}$). The dotted line is constant k_{off}/L; the star marks the default design.

5.4 Choosing the switch temperature

Figure 8 sweeps T_{sw} from 25 to 250 °C. Below about 38 °C the barrier partly switches off during normal operation and the steady peak rises toward the insulator's. Above about 175 °C the barrier switches only after the face of cell 2 has passed self-heating onset, and runaway propagates. Between these limits the switch both cools and protects. The lower edge is set by the operating temperature plus a few switch widths; the upper edge is set by the kinetics.

5.5 Real barrier materials

Each candidate in Table 1 was run at thicknesses from 0.5 to 10 mm in 0.25 mm steps, in both the abuse and the normal-operation scenario. Two numbers summarise a material: the thinnest barrier that confines runaway to the trigger cell, and the normal-operation peak temperature at that thickness (Fig. 9, Table 5).

No candidate contains runaway at 2 mm. Even the aerogel, whose room-temperature k is below that of the idealised insulator, lets runaway through at 2 mm. Its conductivity triples between 200 and 600 °C, the range the barrier's hot face reaches, so its effective hot-state conductance at 2 mm is about $40\,\mathrm{W\,m^{-2}K^{-1}}$, above the containment threshold of Section 5.3. What matters for containment is a barrier's conductivity at runaway temperatures, which datasheets often report only incompletely.

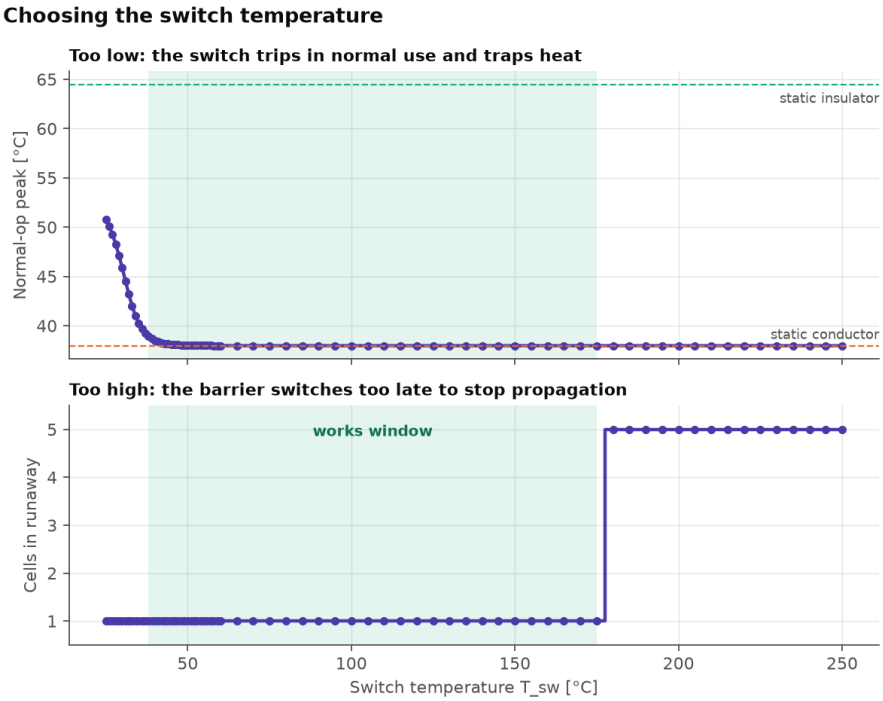


Figure 8: Normal-operation peak (top) and cells in runaway (bottom) as functions of the switch temperature. The shaded window both cools and protects.

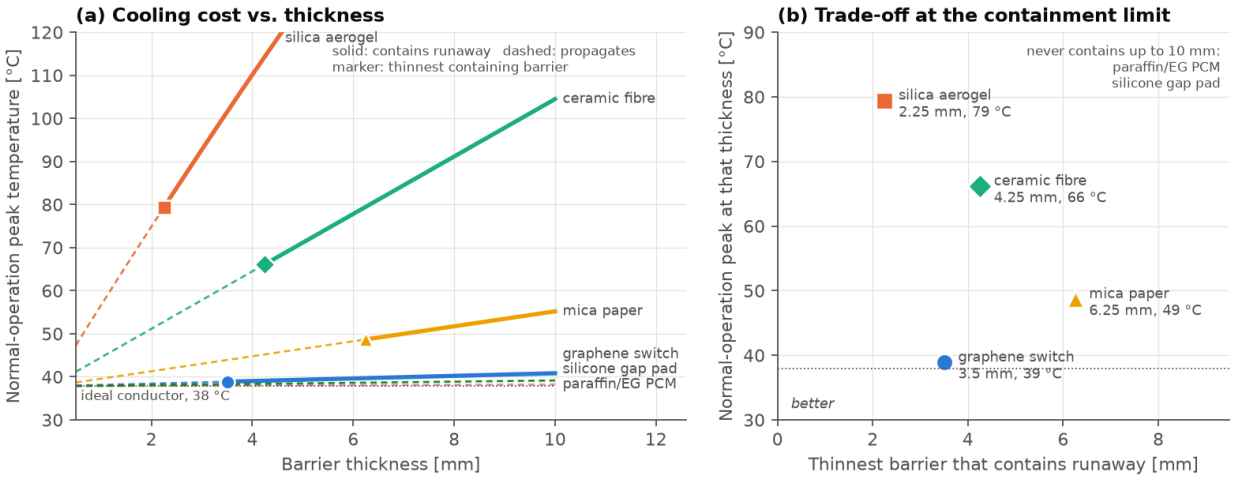


Figure 9: Real barrier materials. (a) Normal-operation peak temperature against barrier thickness; the line is solid where the barrier contains runaway and dashed where it propagates, and the marker shows the thinnest containing barrier. (b) Each material's thinnest containing barrier and its normal-operation peak at that thickness. The lower left is better. The dotted line is the ideal $5\,\mathrm{W\,m^{-1}K^{-1}}$ conductor at 2 mm.

Table 5: Material comparison. "Never" means runaway reached at least one neighbour at every thickness up to 10 mm. At 2 mm all six let runaway reach all five cells.

Material	Thinnest containing barrier [mm]	Normal peak there [°C]	Normal peak at 2 mm [°C]
Graphene switch	3.5	38.9	38.4
Silica aerogel	2.25	79.4	74.9
Ceramic fibre	4.25	66.1	51.1
Mica paper	6.25	48.7	41.3
Paraffin/EG PCM	never	–	37.9
Silicone gap pad	never	–	38.1

Static insulators trade thickness against normal-use temperature. Aerogel contains runaway with the thinnest layer, 2.25 mm, but raises the normal-operation peak by 41 K over the conductor. Ceramic fibre needs 4.25 mm and costs 28 K. Mica paper conducts heat about ten times better than aerogel, so its normal-use penalty is small (11 K), but it needs 6.25 mm before its conductance falls below the threshold. For all three the penalty grows steadily with thickness (Fig. 9a).

The conductor and the phase-change composite fail. The gap pad behaves like the idealised conductor and never contains runaway. The PCM also conducts well ($k \approx 10\,\mathrm{W\,m^{-1}K^{-1}}$), so it keeps cells cool, and its latent heat does slow propagation: at 4 mm, runaway reaches cell 5 after 148 s instead of 100 s without latent heat. But a 4 mm layer stores only about $1\,\mathrm{MJ\,m^{-2}}$ of latent heat, against about $15\,\mathrm{MJ\,m^{-2}}$ of reaction heat released by each cell, and once melted it passes heat on readily. Only at 9.5–10 mm does it hold runaway to two cells.

The thermal switch avoids the trade-off. The graphene switch contains runaway at 3.5 mm while keeping the normal-operation peak at 38.9 °C, within 1 K of the ideal conductor. Its hot-state conductance at that thickness, $0.10/0.0035 \approx 29\,\mathrm{W\,m^{-2}K^{-1}}$, is just below the threshold found with the idealised switch, so the design-map criterion carries over to a real material. Because its cold-state conductivity is only $1.33\,\mathrm{W\,m^{-1}K^{-1}}$, the normal-use penalty does grow with thickness, but slowly (41 °C at 10 mm).

Sensitivity. The switch data come from a paper abstract, so its density, heat capacity, switch temperature and response time are uncertain. Table 6 varies each one. Across these cases the thinnest containing switch ranges from 3.25 to 5 mm, with the normal-operation peak between 38.8 and 39.3 °C. The result is most sensitive to the hot-state conductivity, as the k_{off}/L criterion predicts, and to a slow response: with $\tau = 30$ s the switch needs 4.5 mm. Reversibility makes no difference here, because the barrier next to the runaway cell does not cool below T_{sw} during the run. Doubling the switch's assumed heat capacity, or cutting the aerogel's by 30 %, changes its thinnest containing barrier by at most 0.25 mm.

Which material is best? Judged on both criteria together, the graphene switch is the best of the six. It is the only candidate that contains runaway at a practical thickness without a normal-use penalty, and it stays the best in every sensitivity case. Aerogel is the choice if thickness is the only constraint and the pack can remove the extra heat another way, for example with side cooling, which this 1D model cannot represent. Among commercially established products, mica paper is the best compromise if about 6 mm per gap and a normal-use peak near 49 °C are acceptable. The switch

Table 6: Sensitivity of the thinnest containing barrier to inputs the data do not pin down. Baseline: graphene switch 3.5 mm, aerogel 2.25 mm.

Case	Thinnest containing [mm]	Normal peak there [°C]
Switch, $T_{sw} = 70\,°C$	3.75	38.9
Switch, $T_{sw} = 100\,°C$	3.5	38.9
Switch, instantaneous ($\tau = 0$)	3.25	38.8
Switch, slow ($\tau = 30\,s$)	4.5	39.2
Switch, reversible	3.5	38.9
Switch, $\rho c_p \times 2$	3.25	38.8
Switch, $k_{off} \times 1.5$	5.0	39.3
Aerogel, $\rho c_p \times 0.7$	2.25	79.4
PCM without latent heat	never	–

result rests on a laboratory material characterised here only by its published headline numbers, so measured $k(T)$, heat capacity and cycling data are needed before it can be recommended for a real pack.

5.6 Formulation design

The products above are fixed. This section uses the model of Section 2.6 to ask what a silicone barrier should be made of. Each formulation is scored like the products: the thinnest barrier that confines runaway to the trigger cell, found by bisection to 0.125 mm between 0.25 and 10 mm, and the normal-operation peak at that thickness. The reference materials were rescored the same way. On the finer grid mica contains at 6.125 mm (48.5 °C) and ceramic fibre at 4.125 mm (65.3 °C); the other values match Table 5.

Screen. The screen crossed 21 conductive systems with five microsphere fractions (0–40 vol%) and five endothermic options (none, or 10 or 20 vol% ATH or MDH), keeping total filler at or below 60 vol% as a castability limit (assumed). That gave 452 formulations. The conductive systems were: none; silica-coated silver nanowires of three sizes (40 nm × 40 μm, 60 nm × 20 μm, 100 nm × 15 μm) at 0.5–3 vol%; h-BN at 10 and 20 vol%; 5 μm alumina at 20 and 30 vol%; and four hybrids of 1 or 2 vol% of the thinnest wires with 10 vol% h-BN or 50 nm alumina. Of the 452, 361 contain runaway at some thickness up to 10 mm (Fig. 10).

Static formulations repeat the material trade-off. Without microspheres a formulation is a static barrier, and the k_{off}/L criterion of Section 5.3 applies to its single conductivity. Only 55 of the 105 static formulations contain runaway at all. Neat silicone needs 5.6 mm and peaks at 49.1 °C, close to mica. Filling raises k, which lowers the normal-use peak but raises the thickness needed: the best static formulation (30 vol% alumina + 20 vol% ATH) peaks at 43.1 °C but needs 9.25 mm. Static nanowire formulations contain only at low loadings (0.5–2 vol%, $k \leq 0.54\,W\,m^{-1}K^{-1}$) and then need 6–9.25 mm. Above about $0.5\,W\,m^{-1}K^{-1}$, which includes every h-BN formulation, they conduct well enough that, like the gap pad, they never contain. No choice of particle, nanoparticle or nanowire escapes the conflict, because a static filler cannot tell normal operation from runaway.

Microspheres are the main lever. Adding expandable microspheres turns the composite into a one-way switch, and 306 of the 347 switching formulations contain. Figure 10b shows that the microsphere fraction matters more than the choice of conductive filler. Every formulation on the screen's thickness–temperature Pareto front used the largest fraction screened, 40 vol%:

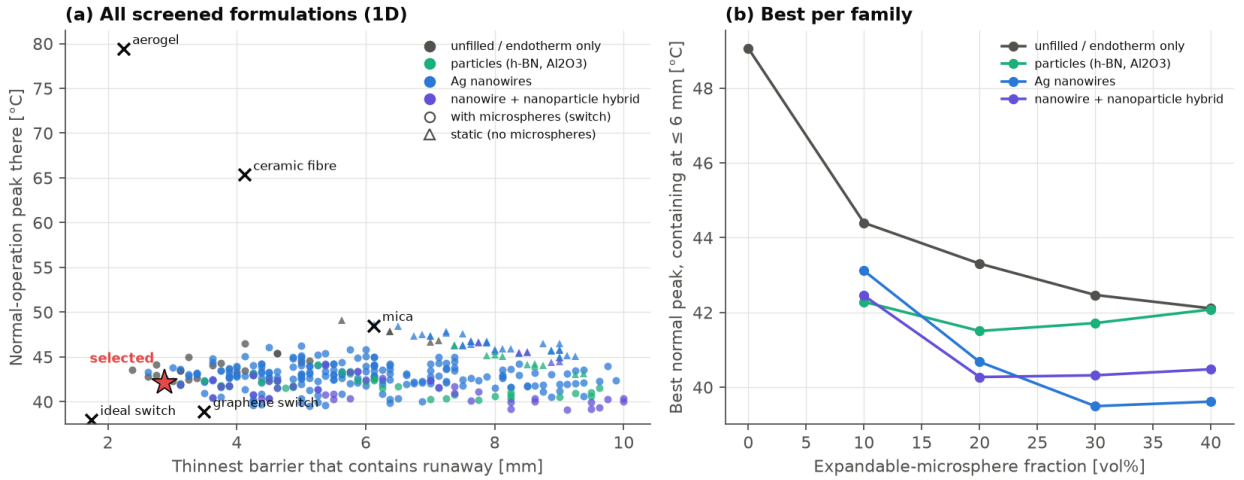


Figure 10: (a) Thinnest containing barrier and normal-operation peak for every screened formulation that contains runaway up to 10 mm (circles: with expandable microspheres; triangles: static), with the reference materials and the selected formulation. (b) The lowest normal-operation peak reached by each family among formulations that contain at 6 mm or less, against microsphere fraction.

Formulation (vol%, rest silicone)	k_{on} / k_{off} [W m^{-1}K^{-1}]	Thinnest [mm]	Peak [°C]
40 microspheres	0.165 / 0.069	2.375	43.6
40 microspheres + 20 ATH	0.268 / 0.093	2.875	42.1
40 microspheres + 10 ATH + 1 AgNW	0.305 / 0.098	3.125	41.9
40 microspheres + 2 AgNW	0.441 / 0.112	3.5	41.0
40 microspheres + 10 ATH + 2 AgNW	0.548 / 0.124	3.625	40.5
40 microspheres + 10 ATH + 3 AgNW	0.943 / 0.157	4.25	39.6

A second pass extended the fraction to 45 and 50 vol%. 50 vol% microspheres with 10 vol% ATH contains at 2.375 mm (42.5 °C), a small further gain at a filler loading that is likely to be hard to process. Figure 11a,b shows why the fraction dominates. Expansion lowers k_{off} roughly in proportion to the gas-cell fraction it creates, while the cold-state k_{on} barely changes. The expansion ratio E matters as much: at 40 vol% the thinnest containing barrier goes from 4.1 mm at $E = 2$ to 1.9 mm at $E = 20$. Even at $E = 2$ (4.1 mm, 44.0 °C) it needs less than half the thickness of the best static formulation at a similar peak (9.25 mm, 43.1 °C).

Nanowires help only if expansion breaks their network. Nanowires are the most efficient way to raise k_{on}, and the best formulations in the screen contained them: 3 vol% wires with 30 vol% microspheres and 20 vol% ATH peaks at 39.5 °C, contained at 5.1 mm. Their weakness is the hot state. The wires that raise k_{on} also raise k_{off}, unless expansion pushes them below the percolation threshold. With 40 nm × 40 μm wires the threshold is only 0.05 vol%, so a five-fold expansion of 40 vol% microspheres ($X = 2.6$) leaves 1–3 vol% loadings far above it. Figure 11c shows the consequence. With weak junctions ($g \leq 0.05$) and a dilute network, wires cost a little thickness and buy a little temperature. With strong junctions ($g = 0.5$) the thinnest containing barrier grows to 5.9 mm at 2 vol% and 8.4 mm at 3 vol%. If the network also survives in the polymer struts, 1 vol% of strongly joined wires, or 3 vol% of weakly joined ones, never contains. The formulation model cannot say which case real composites fall in. Measured nanowire composites show weak junctions (Section 2.6), but these were dispersed in a rigid epoxy and never expanded. Hybrids with nanoparticles did not improve on wires alone in the screen. Nanoparticles raise k_{on} slightly through

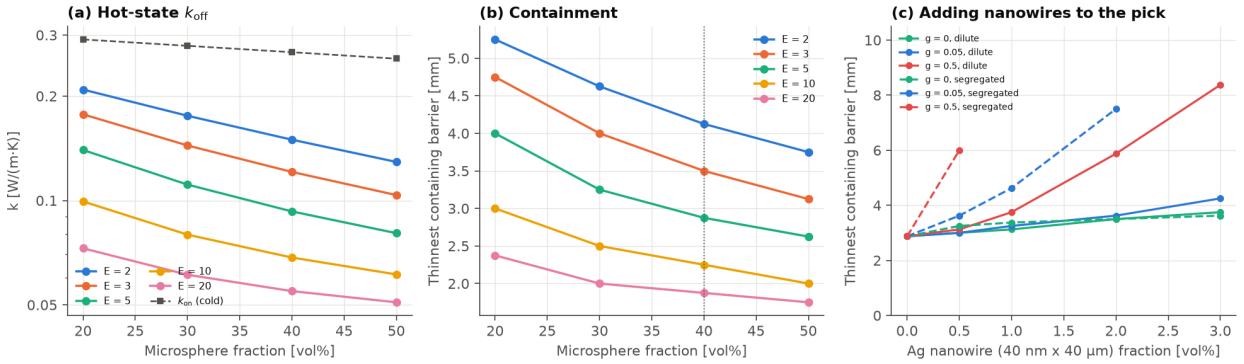


Figure 11: Around the selected formulation (silicone + microspheres + 20 vol% ATH). (a) Hot-state conductivity after expansion against microsphere fraction for several expansion ratios E; the dashed line is the cold-state k_{on}. (b) Thinnest containing barrier. (c) Thinnest containing barrier when 40 nm × 40 μm silver nanowires are added, for junction coefficients g and the dilute (solid) or segregated (dashed) network assumption; missing points do not contain up to 10 mm.

the host conductivity, but they also stay in the hot state and raise k_{off}, and the model credits them with little of the particle-bridging synergy seen in experiments [44].

Endothermic fillers. ATH and MDH absorb 0.4–0.5 MJ kg^{-1} of the composite at 20 vol%, but in this model that does little for containment. At 40 vol% microspheres, adding 20 vol% ATH lengthens the thinnest containing barrier from 2.375 to 2.875 mm, because the particles ($k_p = 3$ W m^{-1}K^{-1}) raise k_{off}. It also lowers the normal-use peak by 1.5 K, because they raise k_{on}. ATH matches or beats MDH in all 139 pairings that contain (2.875 vs. 3.0 mm at 20 vol%), since it decomposes earlier (260 vs. 365 °C). The main reason to keep it is outside the model: ATH is a flame retardant whose water release protects the silicone, which this model does not represent.

Selection. The final choice weighs worst cases, not baselines. The six Pareto-front formulations were rerun under every input the data leave open: the expansion ratio ($E = 3$, 10), a segregated network, the gas-cell conductivity (0.06 W m^{-1}K^{-1}), a higher-temperature microsphere grade (Expancel 920, expanding at 122–206 °C [48]), switch response ($\tau = 0$, 30 s), heat capacity (×0.7) and, for nanowires, junction strength ($g = 0$, 0.5, sintered), poor dispersion and halved wire conductivity. The criterion is the lowest worst-case normal-use peak among formulations whose worst-case containing thickness is at most 4 mm, roughly that of the graphene switch:

Formulation (vol%)	Worst-case thinnest [mm]	Worst-case peak there [°C]
40 microspheres	3.0	45.1
40 microspheres + 20 ATH	**3.5**	**43.0**
40 microspheres + 10 ATH + 1 AgNW	4.0	43.1
40 microspheres + 2 AgNW	5.875	43.7
40 microspheres + 10 ATH + 2 AgNW	6.5	43.0
40 microspheres + 10 ATH + 3 AgNW	9.375	43.0

The 1 vol% nanowire variant is almost tied with the selected formulation, but it adds silver, a dispersion step and the junction uncertainty for 0.1 K. The pick is therefore:

Recommended formulation: silicone elastomer matrix (40 vol%) + Expancel 051 DU 40 expandable microspheres (40 vol%) + 1 μm $Al(OH)_3$ (20 vol%). No conductive filler; electrically insulating.

$k_{on} = 0.27$, $k_{off} = 0.093\,W\,m^{-1}K^{-1}$ ($E = 5$), $\rho = 1320\,kg\,m^{-3}$, $c_p = 1310\,J\,kg^{-1}K^{-1}$, switch at 127 °C (one-way); contains runaway at 2.9 mm with a 42.1 °C normal-use peak; 2.25–3.5 mm and 41.2–43.0 °C across all sensitivity cases.

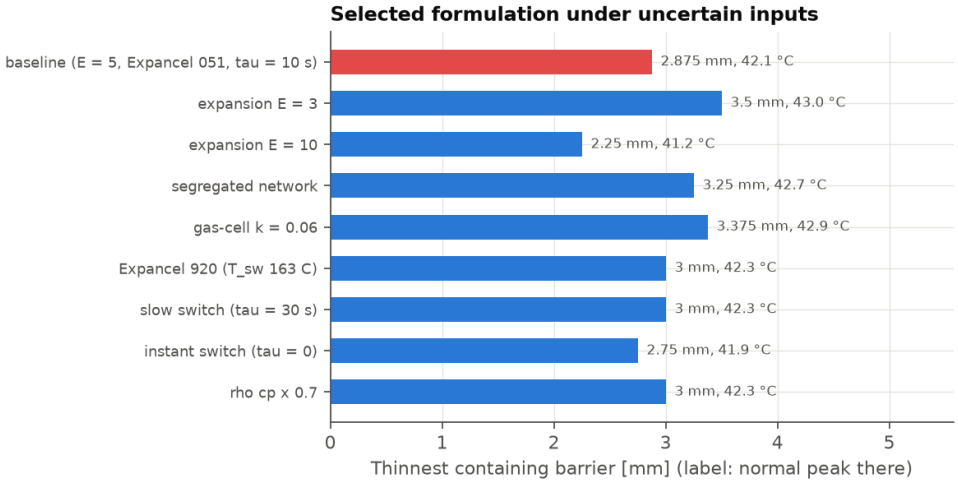


Figure 12: The recommended formulation under each uncertain input: thinnest containing barrier and the normal-operation peak at that thickness.

Figure 12 shows the robustness. The expansion ratio and the gas-cell conductivity matter most, and switch timing, microsphere grade and heat capacity least. Against the products of Section 5.5, the formulation contains runaway in a thinner layer than everything except aerogel. It runs 37 K cooler than aerogel and 6 K cooler than mica at less than half mica's thickness. It does not match the graphene switch, which contains at 3.5 mm with a 38.9 °C peak. The graphene switch's cold-state conductivity is five times higher, and a microsphere composite can match that only with a conductive network that expansion reliably breaks. Whether nanowire or graphene networks in an expanding silicone actually break is the key experiment this study points to. If they do, the 2–3 vol% nanowire formulations reach 39.6–40.5 °C at 3.6–4.3 mm. If they do not, the filler-free formulation above is the safe choice.

6 Discussion and limitations

The model is deliberately minimal, and several simplifications bound what it can say.

- **One dimension.** Heat flows only through the stack thickness, and the stack ends are the only heat sink. Real packs also lose heat sideways to cold plates, housings and busbars. This is why the results depend strongly on h. The containment threshold $k_{off}/L \approx h/3$ should be read as a statement about the competition between barrier leakage and heat removal, not as a design rule.
- **Lumped kinetics.** One Arrhenius step with a rate ceiling captures onset and total heat release, but not the sequence of real decomposition reactions (SEI breakdown, anode–electrolyte, cathode decomposition, electrolyte combustion, internal short) [1, 10].

- **Missing transport.** Venting, ejecta, gas flow and flames, which often carry much of the energy in real failures [2], are not modelled, and neither are radiation across gaps or contact resistance.
- **Simplified switching.** The idealised switch is instantaneous and reversible. The real-material model adds a response time and one-way switching, but not hysteresis, degradation over cycles or latent heat of the switch itself.
- **Incomplete material data.** Several heat capacities are assumed, the gap pad and PCM are extrapolated far beyond their rated temperatures, the aerogel and ceramic fibre data are for industrial blankets rather than battery sheets, and the switch is described by two published points. Decomposition, melting leakage and combustion of organic barriers are not modelled.
- **Formulation model.** The fin and network models are mean-field estimates, checked against two measured nanowire composites in epoxy, not silicone. The junction coefficient, the constrained expansion ratio of microspheres in a cured silicone and whether expansion breaks a wire network are not known and are varied instead. The model under-predicts nanoparticle–nanowire synergy, ignores the viscosity limits of high loadings beyond a 60 vol% cap, and treats the expanded state as fixed: thermoplastic shells soften and collapse well below the barrier's hot-face temperature, and the silicone itself degrades above roughly 300 °C. The selected formulation is the one to make and measure ($k(T)$ through expansion, expansion ratio under the stack's compression), not a validated product.
- **Illustrative cell parameters.** The cell and kinetic values are not fitted to data, so temperatures and times are qualitative. The material ranking is more robust than the absolute thicknesses, which scale with the end cooling h.

Natural next steps are to measure $k(T)$ up to runaway temperatures for the leading candidates, especially the thermal switch, including hysteresis and cycling; to fit multi-step kinetics to accelerating-rate calorimetry and calibrate $k_{\max}$ against measured runaway duration; to extend the model to 2D or 3D with lateral cooling, which would reduce the normal-use penalty of insulators; to compare against an independent commercial finite-element model; and to quantify how parameter uncertainty moves the containment boundary.

7 Conclusions

A verified 1D finite-volume model reproduces the barrier conflict between normal-use cooling and abuse containment, and shows that an idealised switchable barrier resolves it. Containment is controlled by the hot-state barrier conductance relative to the heat removal available to the runaway cell. The switch temperature has a broad working window between the normal operating temperature and the onset of self-heating. On the numerical side, mesh-convergence testing exposed a modelling error that a single run would hide: onset-fitted Arrhenius kinetics extrapolated to burning temperatures make propagation times mesh-dependent, and a physically motivated rate ceiling is needed for a converged answer.

Applied to six real barrier materials, the model shows that the conflict is real for static materials. Insulators that contain runaway either raise normal-use temperatures by 28–41 K (aerogel, ceramic fibre) or need a 6 mm layer (mica), and a conductive pad or phase-change composite does not contain runaway at all. A published graphene/microsphere thermal switch is the best candidate: it contains runaway at about 3.5 mm with a normal-use temperature within 1 K of an ideal conductor, a result that holds across all sensitivity cases. Its standing depends on measured high-temperature properties that are not yet available.

A formulation study then asked what a silicone barrier should be made of. With wires treated as fins and wire networks as a percolating junction network, the model reproduces measured silver-nanowire composites to within about 10 %. Per unit volume, nanowires are the most effective way to raise cold-state conductivity, but no static filler, particle, nanoparticle, nanowire or hybrid escapes the containment conflict. Expandable microspheres turn the composite into a one-way switch, and their fraction and expansion ratio dominate everything else. Nanowires help only if expansion breaks their network, and they make containment much worse if it does not. The recommended formulation, silicone with 40 vol% Expancel 051 microspheres and 20 vol% $Al(OH)_3$, contains runaway at 2.9 mm with a 42 °C normal-use peak and stays within 3.5 mm and 43 °C across all sensitivity cases. That makes it the strongest option that can be formulated from commercial ingredients, second only to the laboratory graphene switch. The decisive next measurements are the constrained expansion ratio and whether a nanowire network survives expansion.

Code and data availability. The model, tests and scripts that regenerate every figure and number in this report are available at `https://github.com/adityameenak/switchable-barrier-runaway` under the MIT license. The solver is written in Python using NumPy [49] and SciPy [50].

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